The Impact of the Rise in Energy Prices during the UK Energy Crisis

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November 10, 2026

Abstract

The UK cost of living crisis following the Ukraine war led to household energy prices doubling. This paper presents evidence on consumer reactions to the crisis and the price elasticity of demand of electricity and gas. Using data from 537,759 households from January 2021 to December 2023, I find that electricity and gas prices rise by 75% and 165% respectively. Their price elasticity of demand is estimated to be -0.037 and -0.44 for these fuels. The findings demonstrate considerable heterogeneity, with consumers being three times more price elastic for gas during the peak of the crisis. Additionally, I find that consumers in more energy efficient homes are less price responsive and low income consumers are significantly more responsive. I find that the rise in prices leads to a significant rise in energy debt and also an uptick in the adoption of smart meters and solar panels. The PED for smart meters is 0.8 for electricity and 0.35 for gas, whereas the elasticity of demand for solar panels is very small.

*Email: nk3186@columbia.edu. I would like to thank Robert Metcalfe, Andrew Schein, Yixin Sun, Louise Bernard and everyone at the Centre for Net Zero for all their help, support and suggestions. The Centre for Net Zero Powered by the Octopus Energy Group helped make this research possible by providing access to data. I remain fully responsible for the paper's contents and any mistakes.

1 Introduction

The Russian invasion of Ukraine led to a rapid unprecedented increase in energy prices worldwide, and this impact was especially prevalent across Europe. Governments rapidly deployed a wide range of substantial support for households (HM Treasury, 2022). The UK government subsidized the marginal cost of electricity and gas, the Energy Price Guarantee (EPG), capping the total average household energy bill at £2500 annually, which reduced prices by 40% (Bolton and Stewart, 2025). Additionally they provided lump-sum payments for all households, in the form of the Energy Bills Support Scheme (EBSS) which consisted of a £400 discount on energy bills across Winter 2022-23. They also provided many other lump sum transfers including: providing households in low council brackets a one off payment of £150 in May - September 2022, one off cash payments totaling up to £900 for low-income benefits, £300 for pensioners, and £150 for disability benefit recipients in the Financial Year 2023-24. These payments were an addition to pre-existing support: the winter fuel allowance, given every year to the elderly and the Warm Home Discount, for low income customers, with an additional Cold Weather Payment provided for colder days. Consumers bills rose rapidly and there was a wide range of media coverage, with many articles demonstrating ways British consumers could reduce their energy bills.

Energy costs have become one of the biggest policy issues in the UK, with the 2026 Budget specifically targeting both gas and electricity costs. The government has created further subsidies, eliminated VAT on electricity, and is providing further lump-sum subsidies for low income households, as well as abolishing gas daily standing charges. Energy debt has also become a severe cause for concern. Consumer energy debt is at an all time high at £5 billion, triple the levels before the 2021 Russian invasion of Ukraine and the UK government is now considering wiping off some consumer energy debt caused by the 2022-23 Energy Crisis to prevent further negative externalities (Department for Energy Security and Net Zero and Ofgem, 2026). They have recently announced an Energy Debt Circuit Breaker scheme which will provide relief, help with repayment plans and reform the current system. In 2023, the median British household had a disposable income of £34,500 (Office for National Statistics, 2024). Despite the EPG subsidy, this still led to the average household spending over 7% of their disposable income on energy. The EBSS would have lowered their annual bill to £2,100, dropping to £1,200 for low-income households during the peak of the crisis. This brings the energy bill of low income consumers back to pre-crisis levels. Despite the wide range of measures that were put in place, reg-

ular households struggled to pay their bills, and crime rates rose due to the rise in energy prices (Fetzer, 2023).

1.1 My Findings

In this paper, I leverage the staggered expiry of fixed price energy contracts to detect how consumers react to a sudden increase in prices and produce new evidence to show how British consumers are much more price inelastic compared to their European counterparts. Consumers who had fixed their energy prices faced a sudden increase in their marginal cost of electricity and gas upon their contracts expiring. They could not have anticipated this rise when they signed their contract, since most fixed contracts range from 12-24 month periods. I use a variety of modern DiD estimators, with my primary specification using the Callaway and Sant'Anna (2021) methodology to show that consumers have a price elasticity of demand of -0.044 for gas and -0.037 for electricity.

These reductions in demand display large levels of heterogeneity across expiry cohorts. Households become three times as price elastic for gas demand during the peak of the crisis. For the entire period, I do not find large elasticities, and I create a model to explain why this may be the case for the UK. I find there to be significant dispersion in the price response across different areas in the UK as well as by a household's energy efficiency. Homes with more insulation tend to need less heating and therefore those consumers have less ability to cut back. Households with solar panels do not change their energy usage for electricity but are just as price elastic as other households for gas. Consumers living in apartments or smaller homes and less expensive homes tend to cut back much further. I find that a large part of the mechanism behind reducing gas usage is by concentrating the hours in which heating is turned on, as we see a rise in the concentration of half-hours that gas is used.

I also find significant differences in the rates of energy debt by property value. Pre-crisis, consumers across all property valued owed small levels of energy debt, whereas the crisis increased energy debt very drastically, and this effect was persistent across all consumers but especially for those likely to be low income. A quarter of these households remained in debt for years after and are still struggling to recover. I find that consumers increase their smart meter adoption rates once their bills rise and have an immediate response after the contract change. Moreover, I find that consumers with the highest usage of electricity adopt solar panels much more than low usage consumers during the crisis when prices were highest. I do not see this pattern in other years, suggesting that

solar panel usage rose due to the price rise.

1.2 Relationship to Existing Literature

This paper contributes to the literature in a few ways. First, it builds on the existing studies that also analyze consumer and business responses to expiries of fixed price contracts after the Russo-Ukraine War (Ahlvik et al., 2026; Lunghi et al., 2026; Alpino et al., 2024; Burlinson et al., 2024). We find substantially different results to previous literature, showing that the UK may be an outlier in its demand response. Zapata-Webborn et al. (2024) use counterfactual models trained on historic data and weather patterns in the UK to find consumption changes due to the crisis and find that electricity consumption was 8.4% lower and gas consumption 10.8% lower than the previous winter. They find a price elasticity of 0.10 for electricity and 0.07 for gas consumption. This does not exactly isolate the price effect since they do not have energy contract data, but provides an answer about the overall reduction in energy demand during the crisis. The UK government finds that temperature-adjusted average consumption reduced by 13% and 18% for electricity and gas (Bolton, 2026).

Labandeira et al. (2017) conduct a meta-analysis on the PED of energy demand and find the short run elasticity of demand for electricity to be 0.126 and 0.180 for gas. Many other papers study the change in prices and analyze households' demand responsiveness (Favero and Grossi, 2023; Alberini et al., 2011, 2020; Burke and Yang, 2016; Okajima and Okajima, 2013; Andersen and Dietrich, 2026; Polo and Roccuzzo, 2026). Alberini et al. (2026) find that in Georgia, the price elasticity of demand for gas is only 0.06 to 0.11 when the government makes 2220kWh ($\sim$ 65% of their winter energy demand) free. Table 1 shows some papers which also leverage a similar method to this paper to see how price shocks affected consumption. This paper contributes to this literature by adding in individual customer data and prices for a large sample size with natural gas data. I also have data on energy debt, low carbon technologies, and smart meters which gives us further insight into consumer behavior.

This paper also analyzes smart meter uptake after prices rise, which is an important channel that helped consumers reduce their energy demand. Gans et al. (2013) found that smart meters have been shown to decrease electricity consumption by 11-17% in Northern Ireland, however interestingly Carroll et al. (2014) found that this reduction happens through the channel of the constant reminder and motivation that the smart meter brings, rather than the lack of knowledge about which appliances use electricity.

There is also literature on how the price shock affected other factors. The UK has the highest rates of fuel poverty in Northern Europe, and the NHS is estimated to spend £2.5 billion per year curing illnesses due to cold damp homes (Lee et al., 2022). There are many studies demonstrating how fuel poverty may lead to subsequent mold issues, winter mortality for elders, house dust mite contamination, cardiovascular diseases and also cold and the flu (Whitehead et al., 2022; Sharpe et al., 2015; Champagne et al., 2023) . Heating can also alleviate mental health issues (Sawyer et al., 2022).

Fetzer (2023) found that the EPG provided less generous support for lower energy efficient households and used this to show how burglaries rose by 6-10%. He predicts a scenario with more energy efficient homes or a more equitably distributed EPG subsidy which would have lowered crime rates.

Survey data gives us an insight into British consumer reactions. Huebner et al. (2023) conduct a survey across British households in 2023, finding that the energy crisis led to consumers decreasing their household temperatures by 1 degree celcius to 19.2. They also find that 52.1% of households left some living spaces unheated (up from 35.5% at baseline), and 59.4% reported heating their homes for fewer hours than in previous winters. 12% of respondents could not keep their homes comfortably warm. The most frequent energy-saving behaviors adopted were closing curtains at night, switching off lights, and taking showers instead of baths. The Ministry of Housing, Communities and Local Government (2024) found that 51% of households said they have reduced heating times, 42% used less electricity and 8% said they have stopped heating their homes entirely. Survey evidence therefore suggests that consumers reacted strongly to their price changes and the cost of living crisis as a whole. Matschoss et al. (2025) also find similar responses for Finnish households.

This paper also relates to the literature on energy efficiency. Fuerst et al. (2015) finds the property price premium on more energy efficient houses, implying that consumers recognize the energy savings they would gain from a more energy efficient home, where homes in EPC bands A and B sell for 5% higher than others (£8900). Braakmann et al. (2026) found that the Russian invasion of Ukraine led to property prices increasing by £1000 for the most energy efficiency properties while they fell by £2400 for the least efficient homes. They explain the small magnitudes of these price changes by the fact that consumers expected the high prices of the crisis to be temporary. Ou et al. (2026) also find similar results, showing that the “brown discount" worsened sharply. Fetzer et al. (2022) develop this by showing how the Energy Price Guarantee subsidy disproportionately helped high income consumers, who generally live in less energy efficient homes,

Table 1: Comparison of Empirical Papers on Energy Crisis Price Shocks

Feature / Dimension	**Ahlvik et al. (2026)**	**Lunghi et al. (2026)**	**Alpino et al. (2024)**	**Burlinson et al. (2024)**
Data	Households (Finland)	Households (Italy)	Industrial / Manufacturing Firms (Italy)	Households (Great Britain)
Energy Type	Electricity only	Electricity only	Electricity and Gas	Electricity and Gas
Meter Data	✓	✓	(✓)	✓
Contract Prices	×	✓	✓	✓
Sample Size	270,054	61,845	837	2,980
Demand Elasticity	**Electricity:** −0.184; −0.229 (rural), −0.243 (indebted)	**Electricity:** −0.113	**Electricity:** 0.0; **Gas:** −1.1	**Electricity:** −0.165; **Gas:** −0.185
Other Relevant Contributions	• Earnings (+1.4%) • Bill defaults (+0.4 pp) • General consumption (−4.5%)	• Unpaid bills (+0.3–0.6 pp)		• Extra expenditure (£131/yr elec, £201/yr gas; £332 total)
Limitations	No price data	No gas data	No household data	Households have to opt in to share data

Notes: This table compares empirical papers examining behavioral and consumption adjustments to retail energy price spikes during the 2021–2023 European energy crisis using quasi-experimental contract roll-offs.

and decreased the incentive for them to upgrade their homes and add insulation and other energy saving technology. If they had been incentivized to upgrade their properties, they could have saved 25 to 40 million tons of CO_2 per year.

Finally, this paper speaks to the welfare effects of the Energy Price Guarantee and its effectiveness. Levell et al. (2025) use the Ofgem energy price caps, EPG changes and transaction level bank account data in the UK to show that income and household characteristics explain only 20% of household variation in energy consumption. They find lagged energy usage is the best metric to predict future energy use and show that the PED of all energy demand was 0.33. They demonstrate that the EPG and EBSS were both regressive policies, however do show that the interventions prevented severe energy poverty: households spending >10% of after-housing-costs income on energy would have spiked to 9.6 million (from 3.9 million in 2019). Despite this, there were severe welfare losses and they calculate an marginal value of public funds (MVPF) of 0.88 for both the policies combined.

Furthermore, Moll et al. (2023) show that in Germany, they only subsidized unit rates of gas and electricity up to 80 and 70% of annual historical household baseline consumption respectively. This was set by the government and ensured that consumption incentives were not skewed. Fetzer et al. (2024) create a model to show how this subsidy design for gas, the Gas Price Brake (GPB), was more efficient and equitable than the UK's EPG. They use UK local election data, showing that energy consumption is strongly positively correlated with votes for the Conservative Party and that the government was vote-seeking by creating the EPG rather than alternative policies.

However, this theory was shown to be largely incorrect due to incomplete information. Dertwinkel-Kalt et al. (2024) conducted a field experiment with a German utility company where they they provided consumers with information about Germany's subsidy, to explain that marginal consumption was not subsidized and therefore they should save more energy. They found that consumers did not understand how it worked and did not alter consumption much. They also found low PEDs, which only explained 0.5 - 3.0% of the total 13.4% reduction in energy demand during the crisis. They tried giving consumers rewards for saving energy at different levels (10% and 20%), but found that this did not change demand by much. They found that the reduction in gas consumption was driven by the salience of the crisis and the general social mobilization rather than individual economic incentives. Therefore, they would conclude that the German GPB was no more efficient than the UK EPG. This study may not be externally valid, as consumers had to opt-in to take part in these experiments and they only have 700 households in

their data.

Fabra et al. (2026) find similar results where they demonstrate how consumers in Portugal and Spain were very price unresponsive but reduced demand because they were driven by non-price behavioral energy-saving measures and campaigns. They show that the Iberian policy of subsidizing input prices of gas and coal for generators lowered wholesale prices directly due to the bidding system of the electricity market, and therefore reduced consumer unit rates by 30%. They exported this cheap energy and slightly raised unit rates to pay for the subsidy and therefore created a revenue neutral subsidy which had minimal distortions. The OECD (2023) summarizes a range of policies across the OECD, consolidating the policies which were the most effective. They explain how transport fuel consumption increased in France due to significant cuts in fuel excise taxes, whereas the Netherlands kept their price signals intact and found that consumption fell by over 20–30% across various sectors. They also explain how marginal price subsidizes are fiscally dangerous and vulnerable to the whims of the electricity spot market, and this crisis should be used to learn what energy policies are most effective.

The remainder of the paper is structured as follows. Section 2 provides detail of the empirical framework, Section 3 presents the descriptive statistics and Section 4 presents the results. I present a model in Section 5, Section 6 conducts a welfare analysis of the Energy Price Guarantee and Section 7 concludes, with a discussion for implications on future policy.

2 Background and Empirical Framework

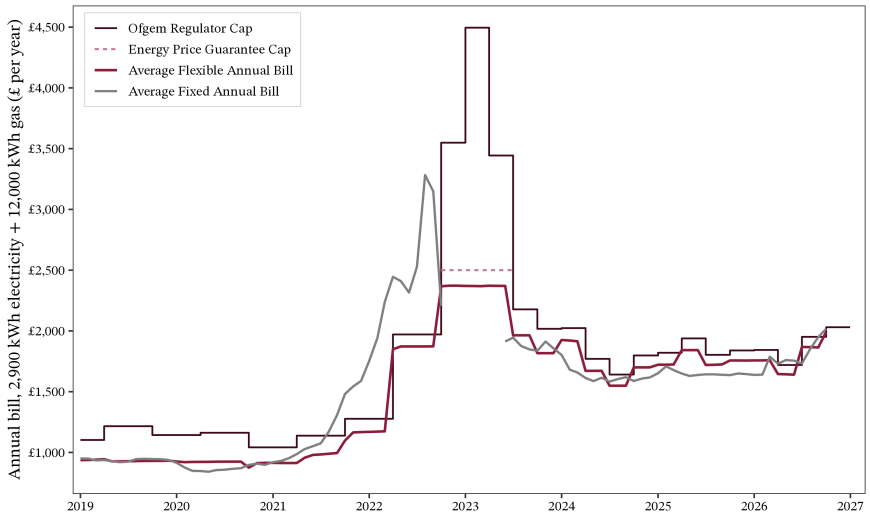
In February 2022, crude oil prices per barrel spiked to over $100, and subsequently electricity and gas prices in the UK followed suit. The UK regulator, the Office of Gas and Electricity Markets (Ofgem), had to rapidly adjust the price suppliers could charge for energy in order to ensure energy suppliers could stay afloat when their input costs rose. The Ofgem price cap incorporates a variety of fixed and variable costs including wholesale energy costs, network costs, operating costs, policy costs, uncertainty, taxes and allows suppliers to retain a small profit margin.

Figure 1 displays the Ofgem price cap from 2019 to 2026 (Office of Gas and Electricity Markets (Ofgem), 2026). The price cap applies to variable tariffs that suppliers charge, and therefore we can see the dark pink line tracks this price cap over time, always staying just below the cap. This is the average price Octopus Energy customers in the UK paid

if they were on a variable tariff each month. The figure also features a gray line which displays the average price of a fixed contract available to customers at each month. This price is set by Octopus to whatever level they see fit, by trying to forecast future energy prices. It begins rising pre-invasion and reaches its peak in mid-2022 when the market predicted that the war was only going to make oil more scarce. From October 2022 to May 2023, Octopus Energy did not provide fixed price contracts to consumers due to the unpredictability of prices in this period.

Additionally in this figure, the pink dashed line displays the Energy Price Guarantee cap at £2500. Consumers paid this lower price while suppliers received the higher price. The difference between these lines is the subsidy the government paid for, which cost the UK over £21 billion.

Figure 1: Annual Bills on Fixed vs Flexible Contract, Overlaid with Ofgem Price Cap and Government Subsidy



Notes: This figure displays total energy bills over time for the average household consumption in 2023: 2900 kWh of electricity and 12,500 kWh of gas. The black line shows the Ofgem price cap over time, the pink dashed line displays the level of the Energy Price Guarantee. The dark pink line shows the prices of an average variable contract and the gray line shows the price of a fixed contract available at that time.

I leverage the sudden rise in energy prices by focusing on consumers who had fixed price contracts during the crisis and so did not face the higher marginal cost of energy until their contracts expired. Prior to February 2022, they could not have anticipated their prices would rise, and all of these consumers signed their fixed contracts during this period. Therefore, a household that signs in March 2021 is not different than one which signs in June 2021. The household who signed in March 2021 will face higher prices a few months before the latter. This makes this a very clean structural design to

detect how consumers react to their prices rising overnight, with no selection bias.

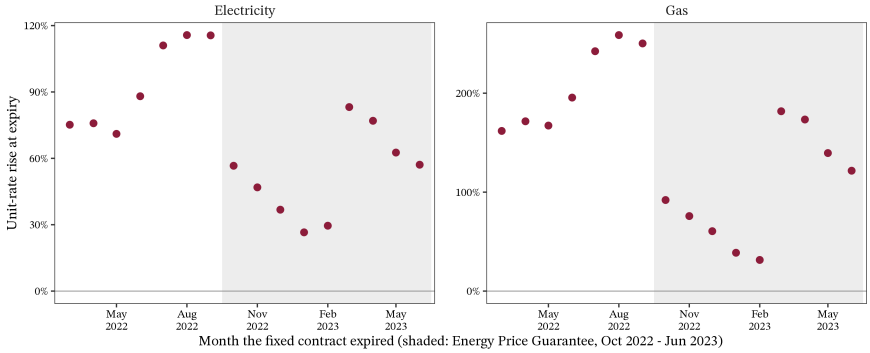
In this paper, treatment is staggered and therefore a two-way fixed effects (TWFE) model here would create biased estimates due to forbidden controls. A TWFE model would compare treated units to already-treated units, and since treatment effects are dynamic and varying across households, this would create a negative bias in our estimates. I expect effects to be dynamic since people adjust their usage over time once they see their energy bills. Treatment effects are also heterogeneous across different types of households, varying by income, energy efficiency, the degree to which they have already cut their usage and their willingness. Therefore, I have used a Callaway and Sant'Anna (2021) estimator to adjust for this. This methods finds the Average Treatment Effect on the Treated (ATT) for each cohort - which groups households by their month of expiry - and conducts a TWFE regression on each cohort and then aggregates this to create a combined event study.

We conduct robustness checks by using other staggered DiD estimators including the Cengiz et al. (2019) stacked design, the Borusyak et al. (2024) imputation estimator and the de Chaisemartin and D'Haultfœuille (2020) strategy.

2.1 Empirical Framework

Figure 2 shows the magnitude of the price rise each of our cohorts faced once their contracts expired. Consumers who's contracts ended immediately after the war began faced an average rise of 75% of their unit rates in electricity and 150% in gas. For consumers ending their fixed price contracts in the peak of the crisis, unit prices of electricity more than doubled whereas marginal rates of gas rose by more than 200%. The magnitudes of these increases were unprecedented and allow this paper to cleanly isolate how consumers reacted.

Figure 2: Price Rise by Expiry Month



Notes: This figure shows the average price rise consumers face in their unit rates of electricity and gas from 6 months before and to 6 months after their contracts expire. The price rise falls in the peak of the crisis due to the EPG subsidy, which those months shaded in gray.

To examine the effect of a sudden price rise, I use several difference-in-differences (DiD) estimators, primarily the Callaway and Sant'Anna (2021) estimator. Since treatment is staggered, a two-way fixed effects DiD regression would create biases in the event study coefficients, since it would compare already-treated households with treated ones. This will bias the event study coefficients if treatment effects are heterogenous across time or across households. Both of these are likely to vary since households will react to price changes differently as they face the higher tariff for longer. In addition, effects are expected to vary according to household characteristics and preferences. Therefore I use the Callaway and Sant'Anna (2021) equation below

$$\widehat{ATT}_c(g,t) = \frac{1}{N^T_{cgt}} \sum_{\substack{i:G_i=g,\\ c(i)=c}} \Big(Y_{it} - Y_{i,g-2}\Big) - \frac{1}{N^C_{cgt}} \sum_{\substack{j:c(j)=c,\\ G_j\neq g}} \Big(Y_{jt} - Y_{j,g-2}\Big), \qquad N^C_{cgt} \geq 200, \tag{1}$$

$$\hat{\theta}(e) = \sum_{(c,g)} \frac{N_{cg}}{\sum_{(c',g')} N_{c'g'}} \widehat{ATT}_c(g, g+e), \qquad \widehat{ATT} = \frac{1}{6}\sum_{e=1}^{6} \hat{\theta}(e). \tag{2}$$

where G_i is the month of contract expiry for each household. Households are indexed i and the event study is monthly where each calendar month-year is indexed t. We only include household-months in the sample if they have enough potential control households in their sample, where each household group c is defined by many dimensions:

1. Whether they have a smart meter.

2. If they move onto a fixed or flexible contract when their initial fixed contract expires.
3. The length of time of their fixed price initial contract.
4. The region they live in (classified into 12 ITL regions across the UK).

This ensures the treatment and control households that are compared with each other are similar and we minimize any composition issues, which helps to ensure parallel trends are held.

The Callaway and Sant'Anna (2021) method calculates the Average Treatment Effect on the Treated (ATT) using a TWFE for each expiry cohort as shown in Equation (1). Since this compares similar households on the same month, we control for the weather, season, the news and other idiosyncrasies. We also ensure the control group is at least a month away from expiring, they are in the same household group c and there is at least 200 of them [1]. We compare treated households with control households that have not expired yet and are not in their anticipation months either. I then aggregate for each month relative to the expiry date to create a combined event study, shown in Equation (2). Finally, I average this over the 6 month post-treatment period in my sample to find the average treatment effect on the treated.

Furthermore, I allow for one month of anticipation (base month = -2), since consumers receive an email warning 45 days prior to their expiry date and may begin decreasing their consumption prior to the price change (Octopus Energy, 2026). Therefore we compare the outcome variable using $Y_{it} - Y_{i,g-2}$.

I use find the price elasticity of demand by finding the Callaway and Sant'Anna (2021) ATT for both consumption and prices.

$$\text{PED}; =; \frac{\widehat{ATT}\log \text{kWh}}{\widehat{ATT}\log \text{unit rate}} \tag{3}$$

3 Data

I analyze monthly data for 537,759 households from January 2021 to December 2023. This rich microdata covers households over the UK who are on fixed price contracts and

[1] I also conduct robustness checks with 100 and 500 minimum controls per cohort and get the same results in Table 6

face a sudden rise in prices when their contracts expire. The dataset includes customer account ID's, their unit and standing rates for each month for both electricity and gas, and the amount of energy they consume.

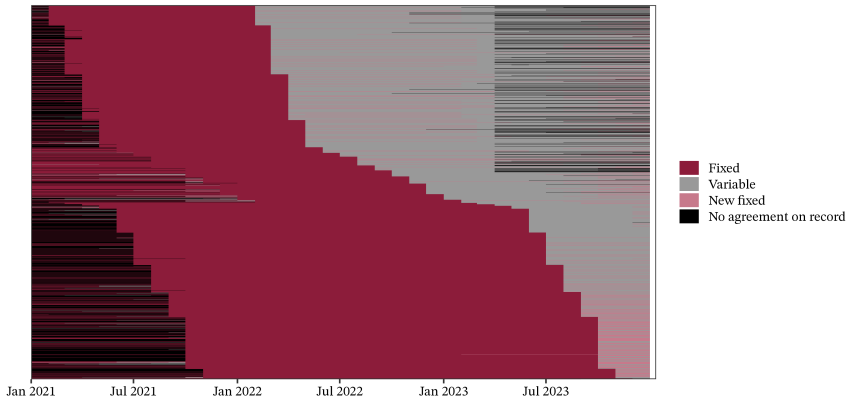
I focus on consumers who end their fixed contracts between March 2022 to April 2023 (196,339 treated households). Figure 2 (figure with how much prices rose for each cohort) shows how prices rose the most rapidly towards the end of 2023. Consumers who ended their contract in March 2022 faced a unit-rate rise of 75% for electricity and 162% for gas, whereas those who's contract ended in September 2022 faced increases of 116% and 251%.

Additionally, I use data on specific household characteristics, including their estimated property value, the year their property was built, its energy efficiency rating, the number of rooms, the floor area, and whether they have any low carbon technologies. I can also estimate when they adopted solar panels, using Octopus Energy installation data, by detecting if they are exporting electricity. Moreover, I use granular information about energy debt that households face over time and how it varies on their payment method.

To ensure I do not retain any data errors, I drop rows where electricity or gas consumption is 0 kWh, as well as months with fewer than 28 days of data. I also drop months where the unit or standing rates are at either extremes and implausible. I drop households with multiple meters for one energy source as well as those who have any consumption above 10,000 kWh in any month. Additionally, I remove accounts that have consumption varying by a factor of 10 between consecutive months and ensure that households have constant rates when they are meant to be on fixed rate contracts. I only keep households who have enough pre-period data and those who have fixes that began before March 2022. Finally, I ensure each household has at least 200 controls which have fixed contracts of the same length.

Figure 3 presents a visual to show how treatment timing is staggered across households. I extract a representative 2000 household sample and show how many are on different contracts and when they are treated. The households at the top of this figure are treated first and the households at the bottom are treated last. The early-treated households are on fixed contracts ranging from 12-24 months, whereas the later-treated households tend to be on 24-month contracts, since they had to be on a fixed contract at the start of the Russia invasion.

Figure 3: Contracts Over Time



Notes: This figure shows how a sample of the customers in my data transition off a fixed contract over time. All of these households signed a fixed contract before March 2022 and each ended somewhere between March 2022 - June 2023.

Additionally, Table 2 provides descriptive statistics about the households in my data. It includes 537,759 households and reports averages over the twelve months before March 2022. These households consume 3,670 kWh of electricity and 14,157 kWh of gas annually, and pay £723 and £550 respectively. This is similar to, if slightly below, the national averages of £766 and £560 in 2021 (Department for Business, Energy & Industrial Strategy, 2022). They owe on average £9 of debt to Octopus Energy, with 10.8% owing at least £5 at some point during the year, and 77% own their own homes. The average home is valued at £350,359 (2026 £), is 101.8 m^2 with 5 rooms, and was built in 1964.

To see if they are comparable, we can test their pre-treatment consumption trajectories, and I find that they are not significantly different, as shown in Table 3. This conducts a regression to see if consumption growth has any relationship to the number of months to the expiry date. The second column controls for the group contract type, which is defined by consumers who have the same type of meter (smart or not), the same contract length and whether they moved onto a fixed flexible contract, as described in Equation (1).

Octopus Energy does not have exact data on who owns an LCT, but we can try and detect who does. I classify households as solar panel owners if they export electricity onto the grid. I can find out if they have electric vehicles and heat pumps if they are on energy saving tariffs for which those technologies are required. I find that in 2026, they own low-carbon technologies at rates similar to the national average, except electric vehicles: 41% have smart meters, 3.2% have solar panels, 7.5% have an electric vehicle and 0.4% have heat pumps.

Table 2: Descriptive Statistics

	Mean	SD	Median
Annual electricity use (kWh)	3,670	2,361	3,137
Annual gas use (kWh)	14,157	7,652	12,964
Annual electricity bill (£)	723	433	621
Annual gas bill (£)	550	280	494
Electricity unit rate (p/kWh)	17.5	2.5	17.0
Gas unit rate (p/kWh)	3.4	1.0	3.0
Electricity standing charge (p/day)	22.5	1.5	22.5
Gas standing charge (p/day)	22.3	2.9	23.8
Debt owed to Octopus (£, monthly mean)	5	52	0
Owed £68 or more at any month-end (%)	5.4	22.5	0.0
Arrears older than 30 days (%)	3.7	18.9	0.0
Owner-occupier (%)	78.0	41.4	–
Property value (£)	347,865	255,865	289,384
Floor area (m^2)	102.1	39.7	92.0
Habitable rooms	5.0	1.6	5.0
Bedrooms	3.1	1.0	3.0
Year built	1964	37	1970
Smart meter (%)	46.6	49.9	–
Households			N = 537,759

Notes: Energy use, bills, rates and debt are averages over March 2021 to February 2022, the twelve months before the first contracts expired; annual use and bills use every month observed in that window, scaled to a year by the share of annual use those months usually make up (the smart-meter extracts begin in April 2021).

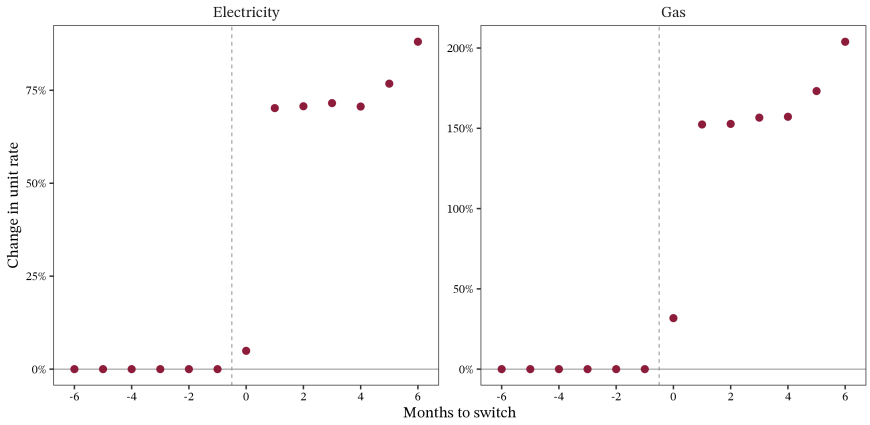
Table 3: Parallel Trends Balance Table

	(1)	(2)
Electricity growth (yoy)	0.032*	0.016
	(0.017)	(0.017)
Gas growth (yoy)	0.026*	−0.001
	(0.014)	(0.015)
Comparison group (sample × fix length × region)	No	Yes
Missing-value indicators	Yes	Yes
Observations	237,545	237,545

4 Results

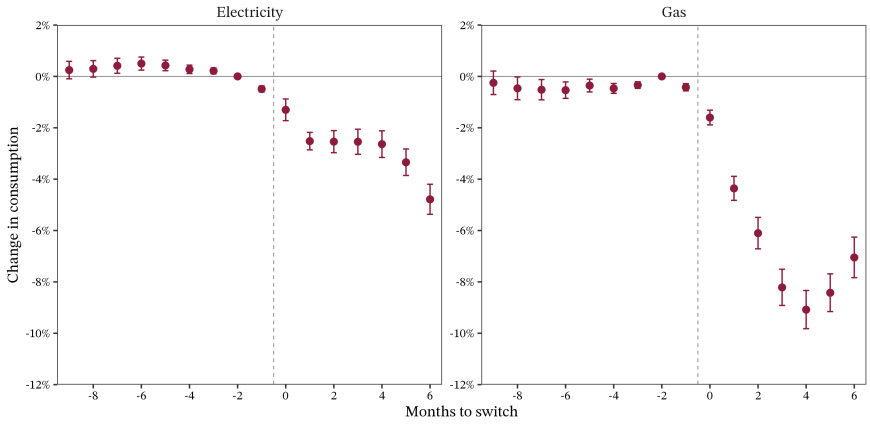
In this section, I present the impact of moving off a fixed contract earlier and facing higher prices sooner. I first look at the impact on prices, energy demand and bills using a Callaway and Sant'Anna (2021) DiD design. Figure 4 displays the impact on prices. There is a clear sudden increase on unit rates by 74.5% for electricity and 165.4% for gas. This is despite the Energy Price Guarantee subsidy which halved unit rates in the peak of the crisis. Using all households in my sample, I find that consumption of gas fell by 7.34% (from 689 kWh per month to 638 kWh) and electricity consumption fell by 2.73% (from 267 to 259 kWh per month), shown in Figure 5. I conduct additional demand change DiD estimators to show that this result is robust to different methods in Table 5. This shows that the consumption change estimate ranges from all of the estimators are very similar. Controlling for weather by finding the Heating Degree Days and Cooling Degree Days does not change the results, as we are only comparing households within similar locations to each other.

Figure 4: Price Rise Event Study



Notes: This figure shows the average price rise consumers face in their unit rates of electricity and gas post-EPG relative to their contract expiry date using the specification shown in Equation (2), without the anticipation (so base month = -1).

Figure 5: Consumption Change Event Study



Notes: This figure shows the average energy demand consumers change relative to their contract expiry date using the specification shown in Equation (2).

The electricity bill for consumers rose by 75.0% whereas the gas bill rose by 111.1% (Figure 31, incorporating standing daily rates for both energy sources. This led to yearly bills rising by £847 (from £961 to £1,808 a year). The pre-trends are flat in both consumption and price. Price is flat by construction, since everyone is on a fixed price contract in the pre-treatment period and therefore their prices will not change. Consumption is flat in the pre-period as we have compared treated households with other control households in the same month, who face the same weather conditions and therefore, since treatment is quasi-random, a later-treated household is no different from an earlier-treated household. We find the overall price elasticity of demand to be -0.037 and -0.044 for electricity and gas respectively. These are significantly smaller magnitudes than previous studies, and also than what various European nations find (Labandeira et al., 2017).

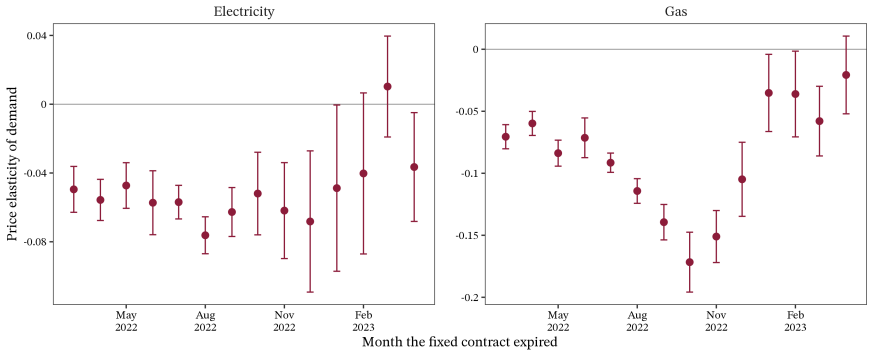
This may be attributed to the cost-of-living crisis. The Russian invasion of Ukraine happened in March 2022, with the peak of the crisis in mid-2022. Many households reduced their energy consumption due to the income effect, as their wallets felt the constraint and the news urged them to reduce their demand. We can see this in Figure 7 where we compare the consumption of electricity between Winter 2021-22 and Winter 2022-23. Many households decreased consumption regardless of the price changed they faced, especially considering that Winter 2022-23 much colder. The Office for Budget Responsibility (2023) find that gas demand fell by 15% adjusting for weather. Leather and Rajham (2026) also find that 79% of households reduce gas consumption by at least 10% and electricity demand by at least 5%, adjusting for temperature.

Households may have already began minimizing their energy consumption and there-

fore when their contract finally expired, they may have struggled to cut further. I create a model that explains why the price elasticity of demand may get smaller in magnitude as consumers get closer to their essential demand levels and can no longer cut. We see evidence for this in Figure 6, where the demand becomes less responsive over time. Earlier-treated cohorts have a PED of approximately -0.04 for electricity and -0.075 for gas, whereas the cohorts that expired in the peak of the crisis have a similar PED for electricity but are much more price elastic for gas (PED = -0.15).

Moreover, households in the UK were provided with significant energy price subsidies. I have already accounted for the EPG subsidy, which significantly decreased unit rates as seen in Figure 1, which decreased the marginal cost of energy and may have nudged households towards consuming more. Additionally, the government provided a wide array of additional lump-sum subsidies to cover energy bills, including £400 over the peak of the crisis, with additional extra funding provided to lower income households, pensioners, and people with disabilities.

Figure 6: Price Elasticity of Demand by Cohort Expiry



Notes: This figure shows price elasticity of demand for each expiry cohort, estimated through the Callaway and Sant'Anna (2021) DiD.

4.1 Cost of Living

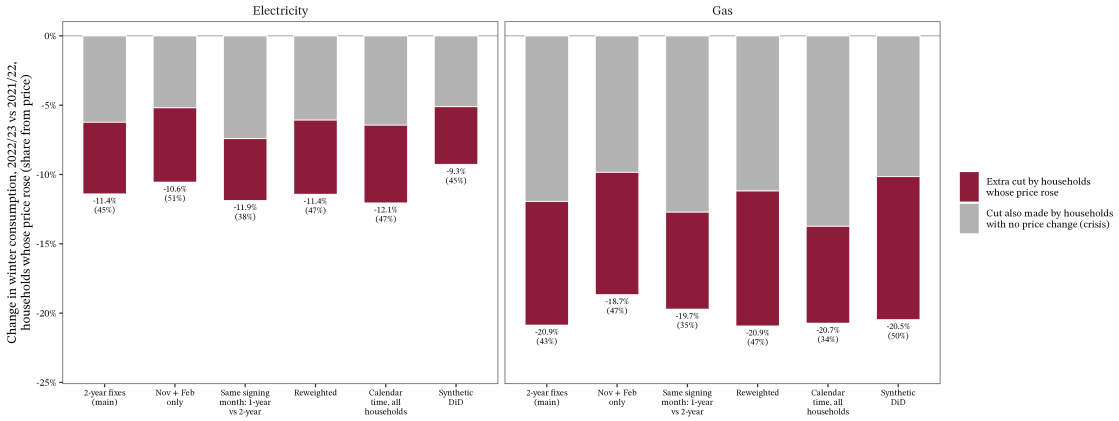
Consumers drastically decreased their energy consumption when energy bills became the largest component of the UK cost of living crisis, being the single largest contributor to headline CPI (HM Treasury, 2023). In this section, I try and quantify the fall from the crisis outside of the price rise. Since all consumers faced this simultaneously, we cannot isolate the causal effect of the crisis entirely. Adjusting for temperature, I find that all consumers decrease their energy usage by around 6.2% for electricity and 12% for gas, similar but smaller to what other studies found (Bolton, 2026). A back-of-the-envelope

calculation suggests that 40-55% of the total demand reduction may have been due to their own prices changes whereas the remaining decrease could be attributed to the cost-of-living effect. To calculate this, I control for temperature, each household's heating sensitivity using past temperature and consumption data, their ownership of smart meters and dropping those with LCTs at any point in our time frame. Nevertheless, this method may still be conflating the impact of consumers adopting other energy efficient technology that I cannot detect. I use multiple methods to try and predict this effect. My methodology is comparing the reduction of winter demand of households who were on fixed price contracts the entire time (the control group in Figure **??** to households who faced the energy shock (the treatment group) from Winter 2021-22 to 2022-23.

Consumers on fixed-price contracts during the entire crisis still decreased their consumption levels beyond what LCTs or insulation changes or other energy efficiency technologies could explain within a year. Figure 7 shows that those on 2 year fixed price contracts through the entire crisis period cut their consumption by 5% and this figure was 11% for those on flexible contracts for electricity (9% and 21% for gas). My previous analysis focuses on the difference between these types of consumers, however it is important to note the impact of the crisis generally, across all British consumers. I also conduct this analysis on different sets of comparison groups:

- Consumers on two year fixed contracts: early vs later expired.
- Comparing consumption across the two winter's in November and February: to avoid Christmas and lockdown period effects.
- Comparing consumers who signed fixed contracts for 1 year with 2 years: the 1 year consumers contracts expired during the crisis whereas the 2 year households were insulated from price effects.
- Reweighted: matching households on winter 2021-22 use, how much their use rose from November to January, region and sample as defined by cohorts in Equation 1
- Calendar time: I regress monthly consumption on each household's own unit rate, with household and month fixed effects. The month effects capture the fall in consumption common to all households (the crisis effect), and the price coefficient captures the additional response to a household's own price rise.
- Synthetic DiD: for each expiry cohort, create a weighted combination of insulated households that have the same consumption levels from November 2021 - September 2022 and then compare them.

Figure 7: Overall Consumption Change Due to the Crisis



Notes: This figure displays a the total change in consumption, and what proportion can be attributed to the price change each household faced.

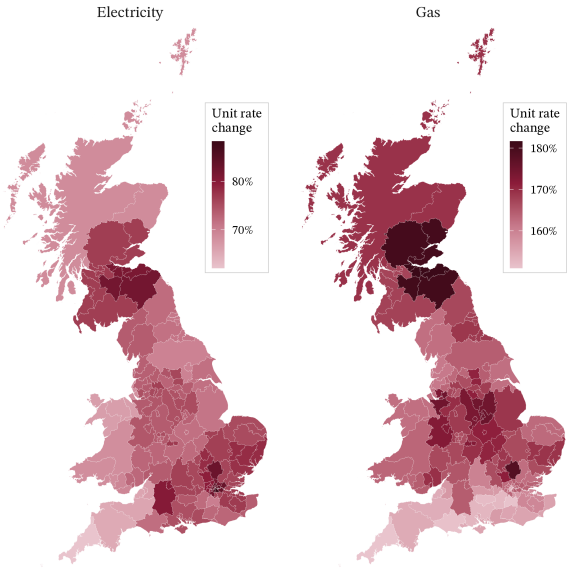
4.2 Heterogeneity

In order to understand household variations in responses to the significant price rise, we conduct heterogeneity analysis across many dimensions of household characteristics.

Firstly, I find that there was significant heterogeneity across the UK. Figure 8 shows how price rises differed across 85 ILT3 regions classified by the Office for National Statistics (2021). For electricity, households in London faced the largest rise in unit rates (82%), followed by the East of England (78%) and Scotland (77%), while Wales and the South West faced the smallest (69%). For gas, Scotland (175%) and the East Midlands (173%) faced the largest rises, and the South East and South West the smallest (157%).

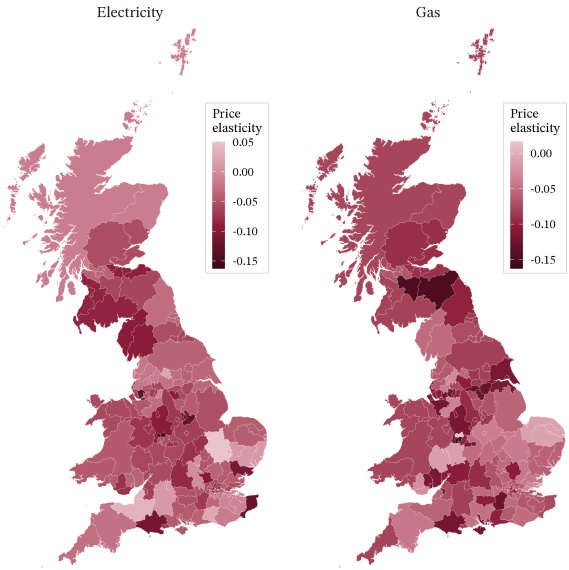
Households in the Midlands were the most responsive to electricity prices (PEDs of −0.078 in the West Midlands and −0.071 in the East Midlands), and those in the South West, Yorkshire and the Humber and the South East the least (around −0.03 to −0.04) and we can see this in Figure 9. For gas, Yorkshire and the Humber (−0.095) and Scotland (−0.088) were the most responsive, and the East of England the least (−0.051). This somewhat correlates with the UK North-South divide, but mostly suggests that regions that responded most with one fuel, substituted to another fuel, for example Yorkshire is among the least responsive for electricity but the most responsive for gas.

Figure 8: Price Rise Map



Notes: This figure displays a map to show regional variation in price rises after contract expiry.

Figure 9: Price Elasticity of Demand by Region

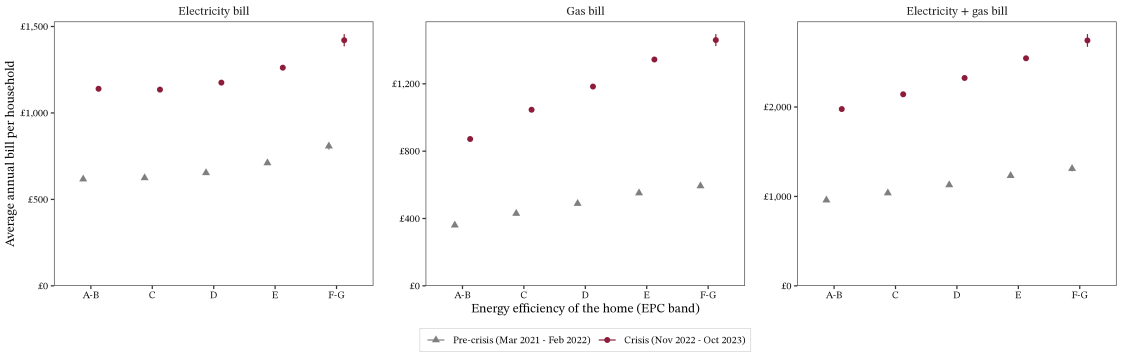


Notes: This figure shows price elasticity of demand by each ILT3 region in the UK.

Across all the heterogeneity analyses I will show below, price rises do not significantly differ by any of these characteristics, and therefore I will discuss the consumption response and elasticity interchangeably, since both effects are scaled versions of each other.

In the UK, households are rated by their home's energy efficiency (EPC). This is based off a range of factors, including insulation, radiators, boilers, windows, cavity walls and other household characteristics that would affect heat loss (Department for Communities and Local Government, 2017). Households across the UK have an average rating of D, where A is the best (very efficient) and G is the worst. Figure 10 shows how consumers with the most efficient homes have the lowest bills.

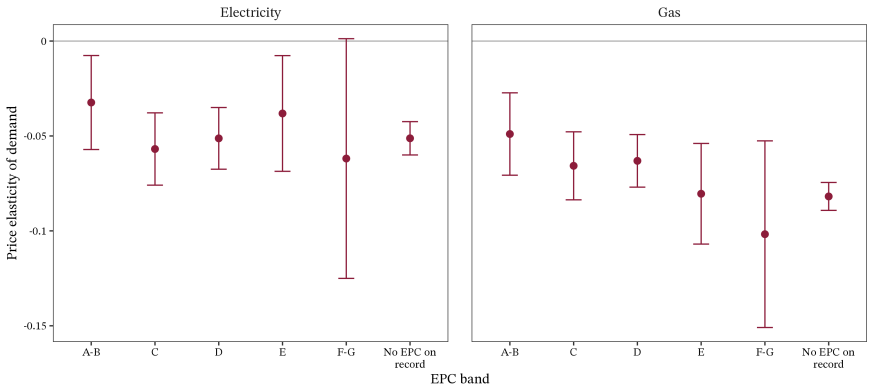
Figure 10: Energy Bills by EPC rating



Notes: This figure displays the average bill of consumers pre and post crisis based on their property's energy efficiency rating.

In response to a major price shock, all households reduce electricity demand at similar levels, which is unsurprising considering the majority 82% of UK households use gas heating (House of Commons Library, 2026). Figure 11 shows that there is a clear gradient with gas demand responses. The most energy inefficient households have the lowest price elasticity, whereas more efficient homes are much more elastic. This suggests that those with more efficient homes are able to reduce their heating and still maintain a comfortable home temperature, whereas those in homes graded A-B cannot reduce their consumption without living less comfortably.

Figure 11: Price Elasticity of Demand by EPC rating

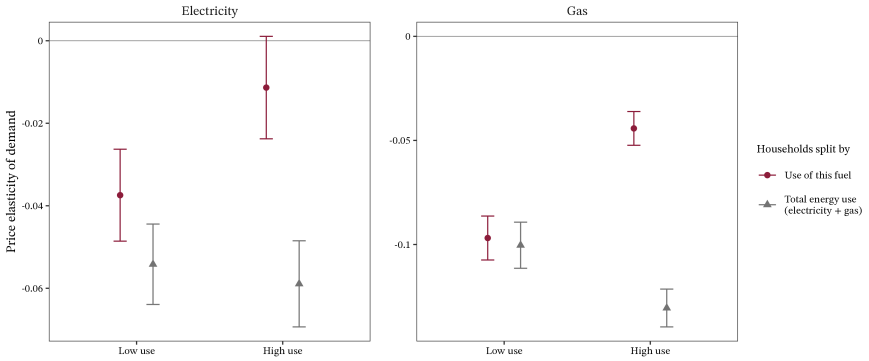


Notes: This figure displays the price elasticity of demand by a household's Energy Performance Certificate rating.

Figure 12 displays the PED and how it varies by energy use. The red circle coefficients show the PED when energy use is defined by each fuel, whereas the gray triangular classify energy use by total usage across electricity and gas. Households who demand below median levels of electricity are more than twice as responsive as those who don't use much electricity. This may potentially point to households who have low carbon technologies and use electricity as an essential fuel and therefore cannot reduce their usage much. This seems corroborated by the fact that households who are high total usage reduce electricity usage a similar amount to those who use less total energy.

Households who use a higher than median level of gas are half as price responsive as households who do not use much gas. This may be related to a lower income or higher household energy efficiency, where low usage households already use less gas but choose to cut further when prices rise. Interestingly, the low usage PED here of -0.097 matches very closely to PED estimations from other papers (Labandeira et al., 2017).

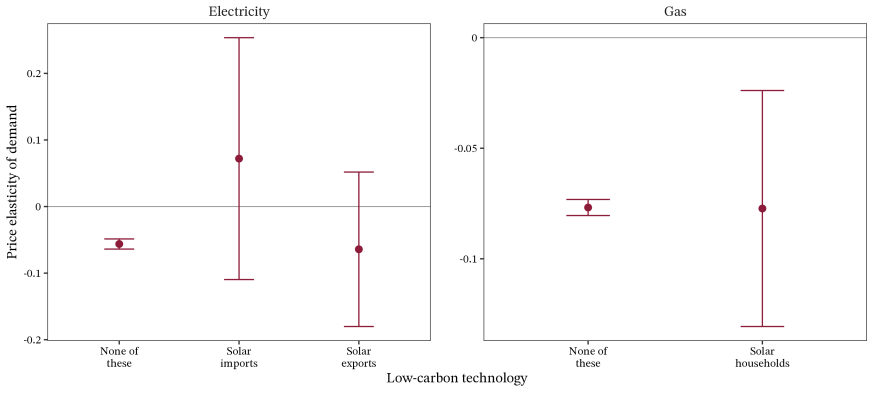
Figure 12: Price Elasticity of Demand by Energy Use



Notes: This figure displays the price elasticity of demand by a household's energy use, classifying them according to households who have below and above median electricity, gas and total consumption.

I look at the response in a rise in prices to energy demand for households who own solar panels. Figure 13 shows that households who do not have solar panels have a PED of -0.05 for electricity. Households who do have solar panels do not change their behavior. They do not need to adjust to energy price changes since they have the potential to export electricity and earn money and do not consume more or less in response to the crisis. Their gas price change response has a similar magnitude to household without solar panels, which is a good robustness check to verify that these households are similar and comparable to other households, with the only difference being their solar panel ownership.

Figure 13: Price Elasticity by Solar Panels

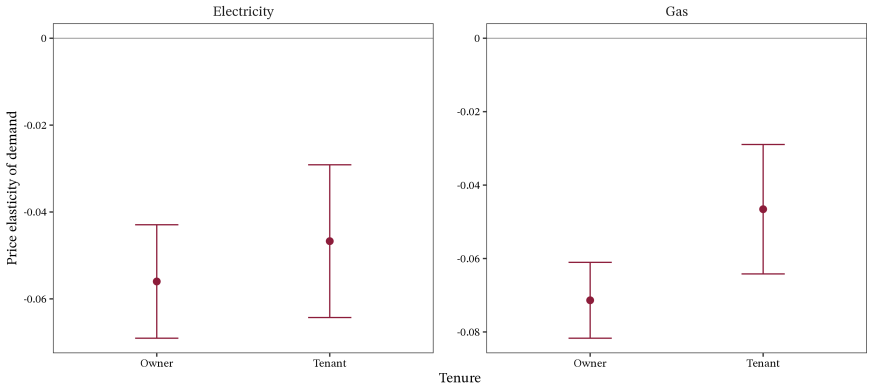


Notes: This figure displays the price elasticity of demand by households who have a solar panel prior to March 2022.

Electricity demand responses to do not differ much according to home ownership, but consumers who own their own home tend to be more price elastic than those who rent

and we can see this in Figure 14. This may be because consumers who rent get their bills included in their contract, or because tenants already minimize heating use and could not cut further when prices rise.

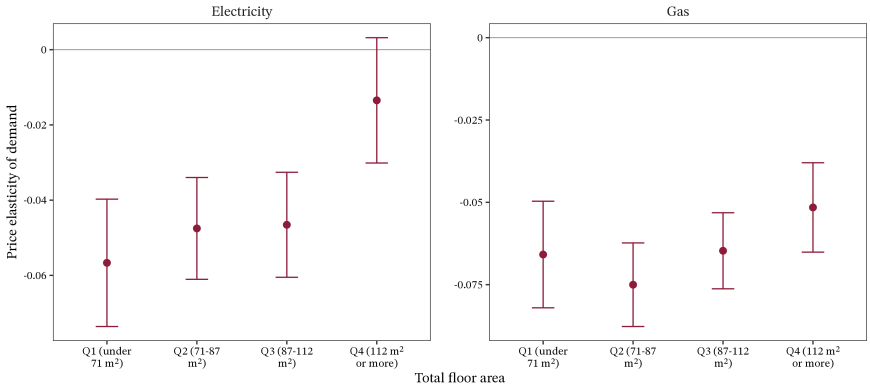
Figure 14: Price Elasticity by Home Ownership



Notes: This figure displays the price elasticity of demand by households according to home ownership.

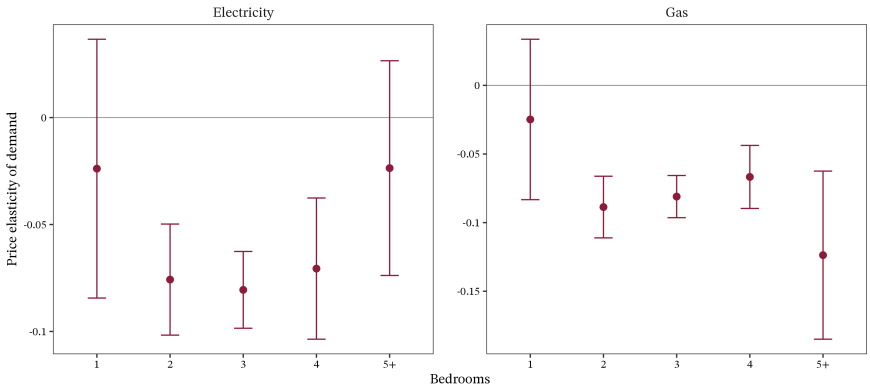
Figure 15 and Figure 19 show the PED by floor area and number of bedrooms and seems to suggest that consumers living in smaller sized houses generally tend to reduce electricity consumption more. However, consumers with more bedrooms tend to be very price elastic (people with 5+ bedrooms have a gas PED of -0.124). This is likely to be due to these households having more room to cut usage. They consume an average of 23,32 6kWh of gas annually, compared to 12,288 kWh for households which have 2–3 bedrooms.

Figure 15: Price Elasticity by Floor Area



Notes: This figure displays the price elasticity of demand by households depending on the size of their home.

Figure 16: Price Elasticity by Number of Bedrooms



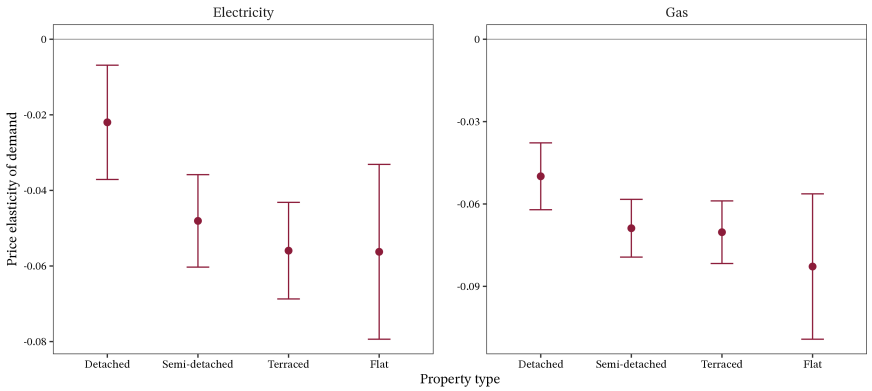
Notes: This figure displays the price elasticity of demand by households depending on the number of bedrooms in their home.

Finally, I look at the type of home that these consumers live in and how that shapes their price responsiveness. Figure 17 shows that consumers with semi-detached, terraced houses and apartments decrease energy usage by more than double compared to those with detached homes. This points to a strong relationship with income, since the majority of homes in the UK are not detached, and the average detached home costs £469,000 - 87.6% more than the others (HM Land Registry, 2026). We see a similar pattern with council tax brackets in Figure 18, where those on lower brackets are more price elastic. Council tax brackets are based off 1991/2003 property values and houses on band A cost less than £40,000 in 1991 whereas those on band G cost more than £320,000 in the same year (CPI adjusted £768,013 in 2026) (Valuation Office Agency and HM Revenue & Customs, 2026; Bank of England, 2026).

To further verify this claim, Figure 20 shows heterogeneity by 2026 estimated property values and finds the same effect. Consumers who reside in more expensive homes tend to be less price elastic than those who live in cheaper homes. This is highly correlated with income, however, unfortunately I do not have income by household so I cannot verify how demand responsiveness to price shocks varies according to income. Consumers who's homes are worth more than £459,000 do have a total pre-crisis annual energy bill of £1,519, compared to £1,008 for those living in homes valued under £181,000. Controlling for the number of bedrooms and floor area, it is still £172 more, however both of those are endogenous to income too, so trying to isolate the income component here is difficult.

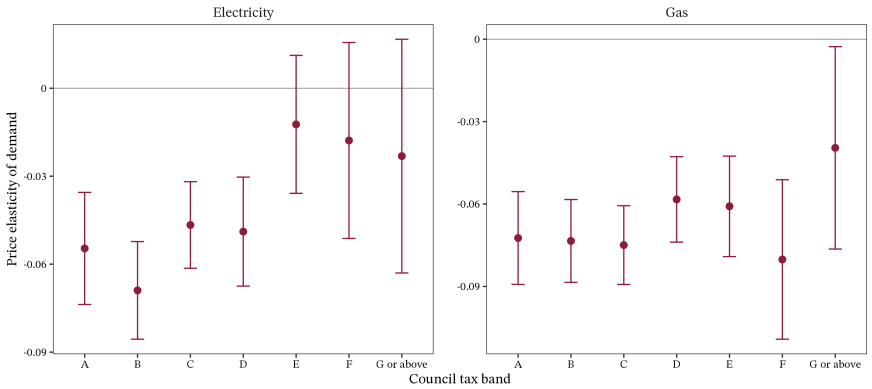
NEED TO FIND OUT WHERE THEY GOT PROPERTY VALUE DATA FROM

Figure 17: Price Elasticity by Property Type



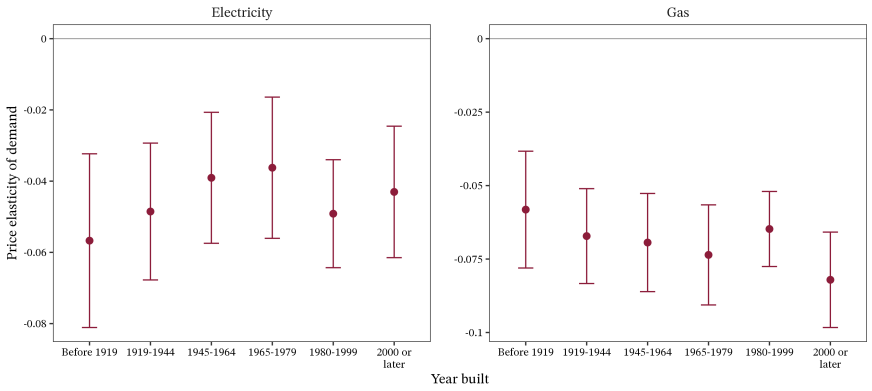
Notes: This figure displays the price elasticity of demand by home type: if it is detached, semi-detached, terraced or a flat (apartment).

Figure 18: Price Elasticity by Council Tax Band



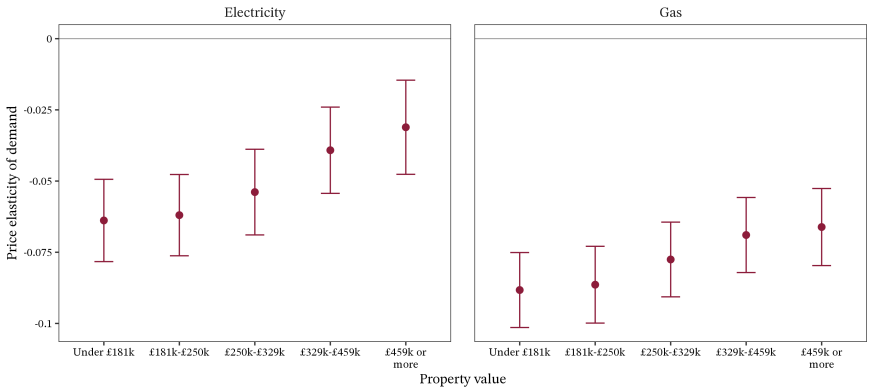
Notes: This figure displays the price elasticity of demand by households according to their council tax band (A = least expensive properties, G = more expensive).

Figure 19: Price Elasticity by Year Built



Notes: This figure displays the price elasticity of demand by households depending on the number of bedrooms in their home.

Figure 20: Price Elasticity by 2026 Estimated Property Values

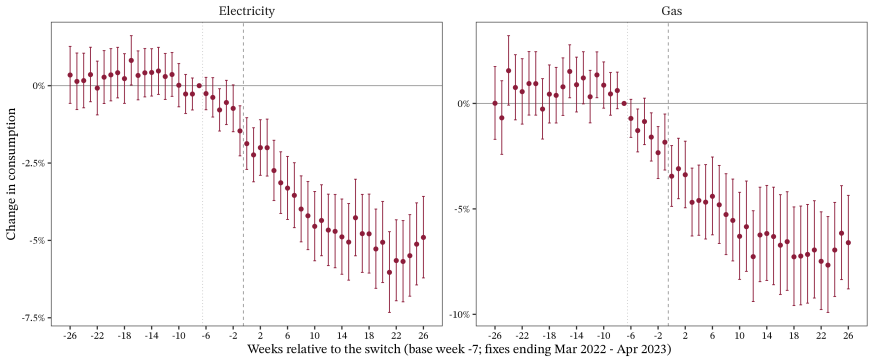


Notes: This figure displays the price elasticity of demand by households according to their estimated property value.

4.3 Granular Data

I test my main specification on consumption data at the weekly level to see how consumers change their demand more granularly. I allow for a 6 week anticipation period, as consumers are given notice of their contracts ending 45 days before it ends, and Figure 21 shows a very clear fall from the anticipation period, which matches the monthly conclusions.

Figure 21: Weekly Consumption Event Study



Notes: This figure displays the Callaway and Sant'Anna (2021) DiD event study at a weekly level for both gas and electricity.

Using half-hourly gas demand data, I construct a Herfindahl–Hirschman Index (HHI) of intraday load concentration to see how the timing of usage changes once customer's face a price rise.

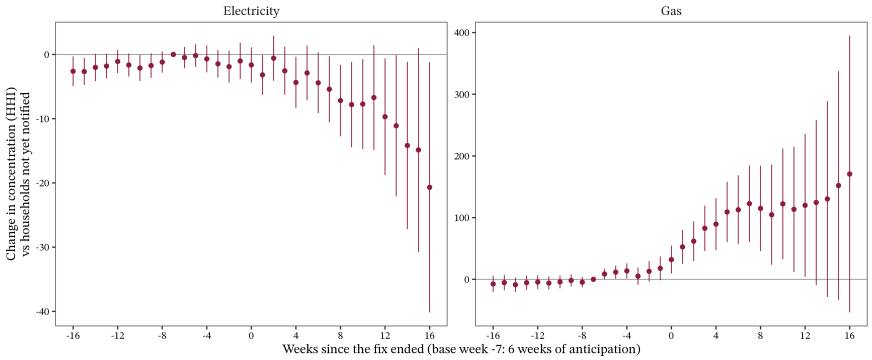
The HHI is calculated as follows:

$$HHI = \sum_{i=1}^{N} s_i^2 = \frac{\sum_{i=1}^{N} x_i^2}{\left(\sum_{i=1}^{N} x_i\right)^2} \tag{4}$$

where x_i is half hourly consumption, N is the number of half-hours in the day (typically 48), and therefore $\sum_{i=1}^{N} x_i$ is total daily consumption and N

Figure 22 shows that there is a very clear increase in the concentration of gas consumptions. Consumers demand less gas overall and also use around 9 kWh of gas every week once their prices rise, and in addition to this they seem to be heating their homes in less time periods through the day and seem to be concentrating their heating into fewer half-hours of the day. This gives us an insight into the mechanism consumers use to reduce their consumption. I do not see this pattern clearly with electricity, if anything they seem to be decreasing their concentration slightly, however this can be explained by the fact that only 6% of households use electricity to heat their homes or for major appliances such as electric vehicles from 2023-2024 (Ministry of Housing, Communities and Local Government, 2025).

Figure 22: HHI Concentration of Usage after Price Change



Notes: This figure displays the Callaway and Sant'Anna (2021) DiD event study for the HHI index of the concentration of usage hours. I use 30,000 random consumers in my sample for this analysis.

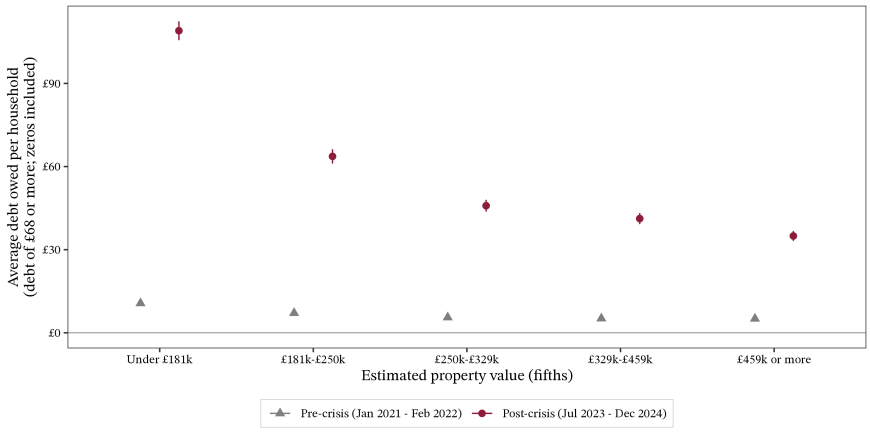
4.4 Impact on Debt

The UK domestic energy debt is severe and recently the UK government has announced a range of measures to tackle it under the 'Energy Debt Circuit Breaker' scheme **??**. Total household energy debt stands at £5 billion and the government attributes this entirely to the price rise from 2022-23. During this period, many energy firms forcibly installed traditional prepayment meters in the homes of those in severe debt to ensure they would have to pay for their energy demand beforehand.

I look at how much average debt rose and how debt levels were distributed among estimated income groups. I also analyze the proportion of customers who were pushed into debt due to the price crisis and have since never recovered and are still struggling to pay their bills.

In my sample, 93.7% of households had never had serious debt before their fix ended. 7.4% of them fell into serious debt for the first time within 12 months after their contract ended. 25% of these households (1.8% of my sample) were still in serious debt in December 2024. Figure 23 shows how much the crisis increased energy debt levels by property value bracket. It is especially concentrated on the lowest income households, showing how the crisis widened inequalities, and despite a range of government interventions to support low income consumers, their debt levels still rose by ten times.

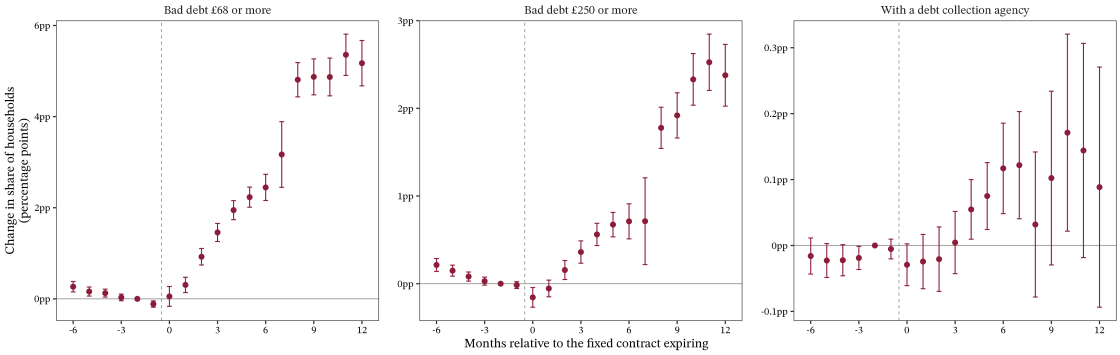
Figure 23: Debt Pre- and Post-Crisis by Property Type



Notes: This figure the average serious debt level of all consumers, categorized by their property value.

Figure 24 shows three measures for the proportion of households in different levels of debt and how the price change at contract expiry leads to a sudden and persistent increase in debt levels. We have evidence of non-parallel trends here, which seems to suggest that consumers who have their contracts expiring soon are making attempts to reduce their debt. Octopus Energy classifies dangerous debt as debt above £68 and I find that the number of households in debt rises by 6 percentage points. Very serious debt of £250 or more rises by 3 percentage points and there is an uptick in the proportion of customers whose debt gets sent to a debt collection agency. The raw numbers for these measures become higher for all consumers, and this figure just isolates the impact of their contract expiry on debt. Many consumers generally faced high debt levels due to the liquidity constraints they faced due to the cost-of-living crisis, however they serve as the control group here and we cannot causally assess how they were impacted.

Figure 24: Debt Changes after Price Rise



Notes: This figure shows a Callaway and Sant'Anna (2021) DiD event study on different measures of debt levels.

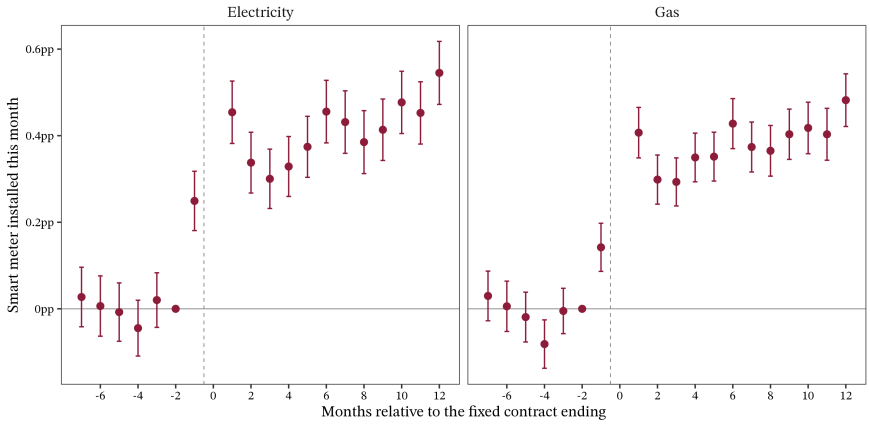
4.5 Impact on Smart Meter Adoption

After facing a large rise in prices, we find evidence that consumers want to track their consumption more carefully and therefore choose to install smart meters. Huebner et al. (2023) found that smart meters served as a critical psychological lever. They found a correlation between how frequently a household checked its consumption on the smart meter and the amount of effort they claimed to exert to reduce consumption.

I find that households adopt more smart meters in months with larger price rises. Figure 25 shows that smart meter adoption rises around 0.4 percentage points every month, which is a 50% rise from the baseline of 0.75% of households per month. This effect persists and one year after expiry, 5% more households have a smart meter. I find that consumers do not change their smart meter adoption patterns when prices do not change much upon contract expiry. The increase in smart meter adoption is most salient when prices rose most at expiry in mid-2022, whereas there is no increase in adoption where the price change at expiry was small or negative.

After their fixed contract ends, households' monthly smart-meter adoption rate rises by about 50%. This implies an elasticity of smart-meter adoption with respect to the energy price of roughly 0.8 for electricity and 0.35 for gas, though since most households install both meters at once, these reflect a single response to the rise in their overall bill. For every 10% rise in energy prices at expiry, consumers are 8% (electricity) and 3.5% (gas) more likely to adopt smart meters in a given month.

Figure 25: Smart Meter Adoption Event Study



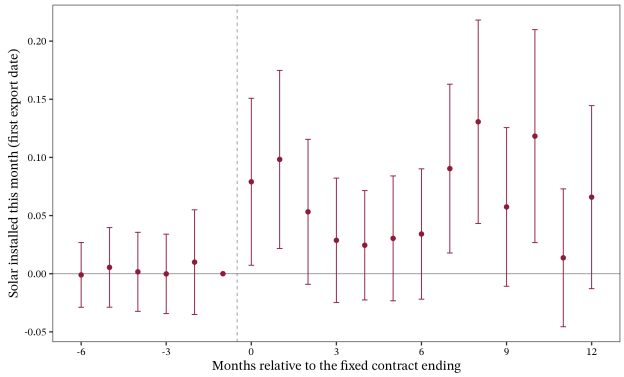
Notes: This figure shows a Callaway and Sant'Anna (2021) DiD event study on smart meter adoption for consumers whose contracts expired March 2022 - April 2022.

4.6 Impact on Solar Panel Adoption

I also investigate whether a sudden price shock to electricity will induce consumers to adopt more solar panels. I find that 0.45% more households buy solar panels once their prices rise, shown in Figure 26. The small affects here may be explained by the fact that solar panel installation costs at least £6,663 and the price shock would create liquidity constraints for households [2]. I also see suggestive evidence in Figure 27 that consumers who have more incentive to install solar panels are more likely to do so during the energy crisis.

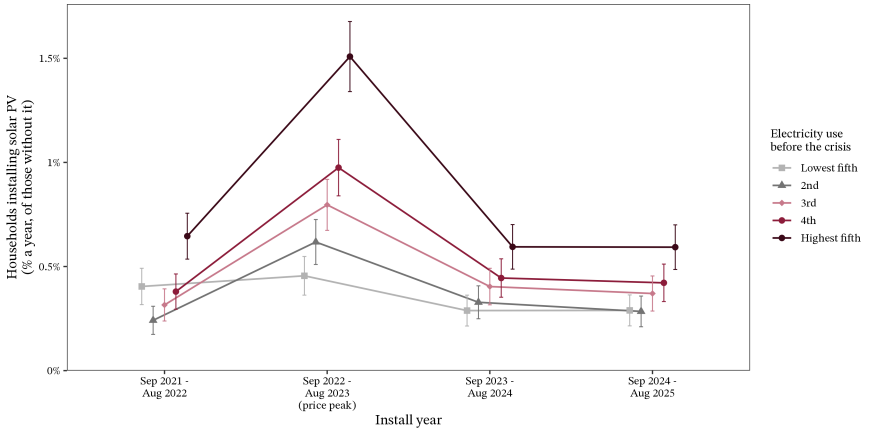
[2] See solar panel installation costs

Figure 26: Solar Panel Adoption Event Study



Notes: This figure shows a Callaway and Sant'Anna (2021) DiD event study on solar panel adoption for consumers whose contracts expired March 2022 - April 2022 and they went from one fixed contract to another fixed contract. This was a necessary filter here, because many households who moved to flexible contracts, did so in order to take advantage of Octopus' time of use tariffs. Therefore, there was selection bias into renewing onto a flexible price contract.

Figure 27: Solar Adoption by Usage



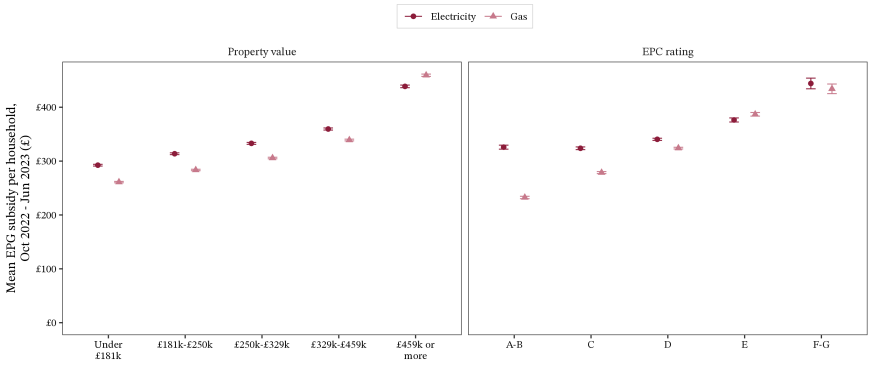
Notes: This figure shows the proportion of households in my data which installed solar panels by year and their pre-crisis usage of electricity.

5 Subsidy and Welfare Implications

The UK government paid for the Energy Price Guarantee, a subsidy on unit rates of electricity and gas, which reduced marginal costs of consumption by around 40% and cost the government a total of £23.6 billion across October 2022 - April 2023 (2.3% of UK government spending in that fiscal year) (Department for Energy Security and Net Zero, 2025).

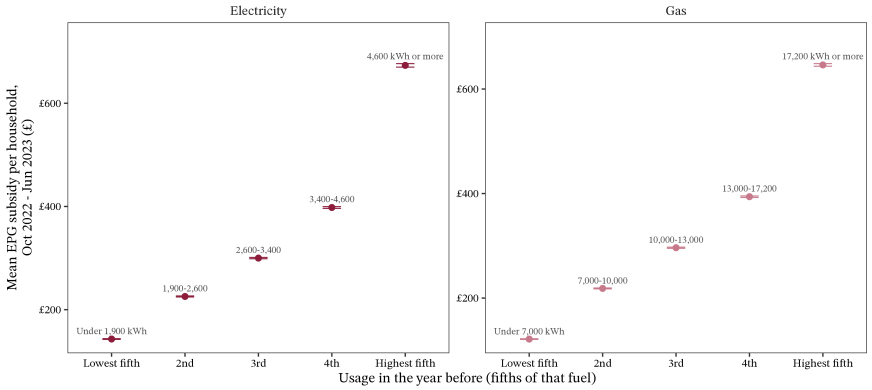
I calculate the average subsidy provided to households in my sample to be £645 and find that this subsidy disproportionately benefited high income, high usage households. Figure 30 shows how consumers living in high value properties receive a much higher subsidy from the government. It also demonstrates how consumers with less energy efficient properties received a higher subsidy, corroborating evidence from Fetzer et al. (2024). Figure 29 emphasizes how unequal and regressive the EPG was. The subsidy impact is also concentrated in London compared to the rest of the UK, as seen in Figure 30. It is clear from these figures that the EPG was not a progressive subsidy and how it led to direct transfers to the most affluent households rather than helping ordinary consumers with the cost-of-living.

Figure 28: EPG Subsidy by Property Value and EPC Rating



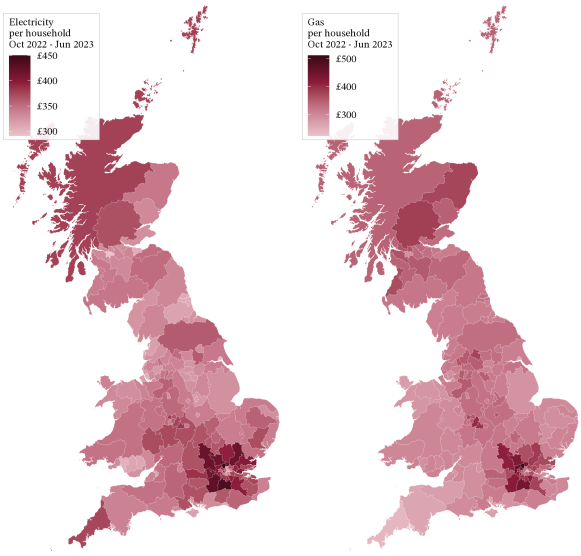
Notes: This figure shows how the EPG subsidy per household varied by property value and energy efficiency (EPC) rating.

Figure 29: EPG Subsidy by Usage Quintile



Notes: This figure shows how the EPG subsidy per household varied by energy usage.

Figure 30: EPG Subsidy by Location



Notes: This figure shows the EPG subsidy per household varied by location.

The EPG subsidy had a range of effects: it reduced competition among suppliers as all utility providers charged the same price for flexible contracts, it led to higher levels of inequality as low income households spent much more of their income on energy bills. Many argue that it discouraged energy conservation and slowed down the green transition as it did distorted household incentives. Consumers may have used less electricity and gas if they had paid the full price, however, the EPG may have also prevented severe fuel poverty rates, household mold, flus and excess winter mortality among other severe diseases by ensuring households heated their homes sufficiently. A flat subsidy may have still led to these issues, since consumers still may have reduced their heating too much and caused long term harm.

The EPG increased fiscal liability and interest rate payments significantly in the long term, however it also reduced inflation (since energy bills are a major component of Consumer Price Inflation) and therefore interest rates did not have to be raised as much, which may have decreased government bond payments. I will not include this effect in my welfare calculation because the magnitudes are hard to assess and the direction of the imapct is not clear.

6 Model

6.1 Part 1: Electricity Consumption, Subsistence Constraints, and Subsidies

6.1.1 Preferences and Baseline Budget Constraint

Households maximize utility, which depends on general consumption C and electricity consumption E. Electricity consumption is bounded below by a strictly necessary subsistence level $\overline{E} > 0$, representing essential heating, refrigeration, and lighting.

Stone-Geary utility (Cobb-Douglas utility function with minimum $\overline{E}$):

$$\begin{aligned} U(C,E) &= C^{1-\alpha}\left(E-\overline{E}\right)^{\alpha} \\ &= (1-\alpha)\ln(C) + \alpha\ln(E-\overline{E}) \end{aligned} \tag{5}$$

where $\alpha \in (0,1)$ is the weight given to extra electricity consumption above the subsistence floor $\overline{E}$.

Let: Y be Household income, p is the price per unit of grid electricity (unit rate of electricity), S is the government lump-sum subsidy payment, and σ is the government subsidy per unit kWh of energy. Therefore households pay effective market price $p_e = p - \sigma$.

Household's budget constraint:

$$C + (p-\sigma)E = Y + S \tag{6}$$

6.1.2 Utility Maximization and Demand Functions

Let extra electricity consumption $\widetilde{E} = E - \overline{E}$. The budget constraint is:

$$C + (p-\sigma)\widetilde{E} = Y + S - (p-\sigma)\overline{E} \tag{7}$$

Solve the Lagrangian for utility optimization: (in appendix)

$$E^* = \overline{E} + \frac{\alpha}{p-\sigma}\left(Y + S - (p-\sigma)\overline{E}\right) \tag{8}$$

6.1.3 Comparative Statics: Price Shocks vs Unit Subsidies vs Lump-Sum Subsidies

To understand how a change in the unit price of electricity and gas changing affects optimal consumption, we differentiate (8) with respect to the price p:

$$\frac{\partial E^*}{\partial p} = \underbrace{-\frac{\alpha(Y+S)}{(p-\sigma)^2}}_{\text{Price Elasticity / Substitution Effect (<0)}} + \underbrace{0}_{\text{Subsistence Floor Constraint}} \tag{9}$$

When p rises:

1. **Price Elasticity Effect:** total electricity consumption E^* falls as consumers try to minimize usage.
2. **The Lower Bound $\overline{E}$:** However, consumers hit a lower bound of minimization. As p rises, extra electricity consumption $(E^* - \overline{E}) \to 0$, so $E \to \overline{E}$ and the price elasticity of demand decreases towards 0. Consumers cannot decrease consumption further even if prices rise more.

To combat this issue and ensure consumers stay above $\overline{E}$, the government created the Energy Price Guarantee Scheme which capped unit rates of electricity and gas.

Effect from unit subsidy σ: To understand how government unit subsidies affect consumption, we differentiate Equation (8) with respect to σ:

$$\frac{\partial E^*}{\partial \sigma} = \frac{\alpha(Y+S)}{(p-\sigma)^2} > 0 \tag{10}$$

The subsidy σ increases energy consumption. It lowers the unit price and chanegs the marginal rate of substitution. (MRS $= p - \sigma < p$). This reduces the price of all consumed electricity, including discretionary consumption above $\overline{E}$. The private marginal benefit ($MB = p - \sigma$) falls below the true marginal social cost ($MSC = p$). This generates over-consumption and deadweight loss (DWL).

Income Effect from Lump-Sum Subsidy S: To understand how government lump sum subsidies affect consumption, we differentiate Equation (8) with respect to S:

$$\frac{\partial E^*}{\partial S} = \frac{\alpha}{(p-\sigma)} > 0 \tag{11}$$

The lump-sum subsidy S increases total disposable income. Because energy has a positive

income elasticity, S creates a positive income effect that directly mitigates the desire to cut electricity consumption.

Therefore, the net change in electricity consumption following an energy price spike p^* accompanied by a lump-sum subsidy S is:

$$\Delta E^* = E^*(p-\sigma, S) - E^*(p_0, 0) = \alpha\left(\frac{Y+S}{p-\sigma} - \frac{Y}{p_0}\right) - \overline{E}\left(\frac{\alpha(p-\sigma-p_0)}{p-\sigma}\right) \tag{12}$$

The combination of mandatory subsistence $\overline{E}$, the EPG subsidy σ and the income effect from S explains why real-world household energy consumption exhibits high rigidity and fails to decline significantly despite steep rate increases.

Ideally the government would provide a lump-sum subsidy $S = p\overline{E}$ targeted directly at covering subsistence energy costs. The demand function becomes:

$$E^*_{\text{lump}} = \overline{E} + \frac{\alpha}{p}\left[Y + \underbrace{p\overline{E}}_{S} - p\overline{E}\right] = \overline{E} + \frac{\alpha Y}{p} \tag{13}$$

This ensures consumers face the true marginal price p, the marginal condition $MB = MSC = p$ holds. The subsidy $S = p\overline{E}$ would purely be an income effect, neutralizing the financial burden of $\overline{E}$ without distorting marginal consumption choices above $\overline{E}$.

6.2 Impact on Solar Panel Adoption

We extend the model to where consumers maximize utility over two time periods ($t = 1, 2$) with a discount factor $\beta \in (0, 1)$. At $t = 1$, an exogenous energy price shock increases grid price from p_0 to p_1.

Let:

- $I \in \{0, 1\}$ denote the binary solar adoption decision at $t = 1$.
- K be the fixed capital investment cost of solar panels.
- $\theta \in (0, 1)$ be the fraction of grid electricity offset by solar generation. If $\theta = 1$, consumers no longer have to pay for electricity.
- ϕ represent search, transaction, and installation friction costs.

The representative consumer maximizes total utility $V = U(C_1, E_1) + \beta U(C_2, E_2)$.

Consumers can save money by adopting solar panels.

$$\text{Savings} = \theta(p - \sigma)E^* \tag{14}$$

Households may consider adopting solar panels in period 1, which will cost K:

$$C_{1,\text{adopt}} = Y + S - (p_1^* - \sigma_1)(1 - \theta)E_1^* - K - \phi \tag{15}$$

The private utility change from adopting in Period 1 relative to non-adoption ($I = 0$) is:

$$\Delta U_1 \approx \underbrace{\theta(p_1^* - \sigma_1)E_1^*}_{\text{Suppressed Bill Savings}} - \underbrace{[u'(C_1) \cdot (K + \phi)]}_{\text{Amplified Utility Cost of Liquid Capital}} \tag{16}$$

The reduced unit rate of electricity reduces the incentive to invest in solar panels, since the potential savings fall. Additionally, the cost of living crisis reduces overall consumption and increases marginal utility of consumption so consumers would prefer to spend more on other consumption in this period rather than investing in solar panels.

6.2.1 Medium-Run ($t = 2$)

In period 2, the EPG from the government goes away, so $\sigma = 0$.

The return on solar panels returns to its full, un-distorted level. Since prices are high, consumers now have incentive to invest in solar energy:

$$\text{Savings}_2 = \theta\, p_2^* E_2^* > \theta(p_1^* - \sigma_1)E_1^* \tag{17}$$

This explains why solar panel adoption fell at the start of the crisis and did not rise as much as we would have expected.

7 Conclusion

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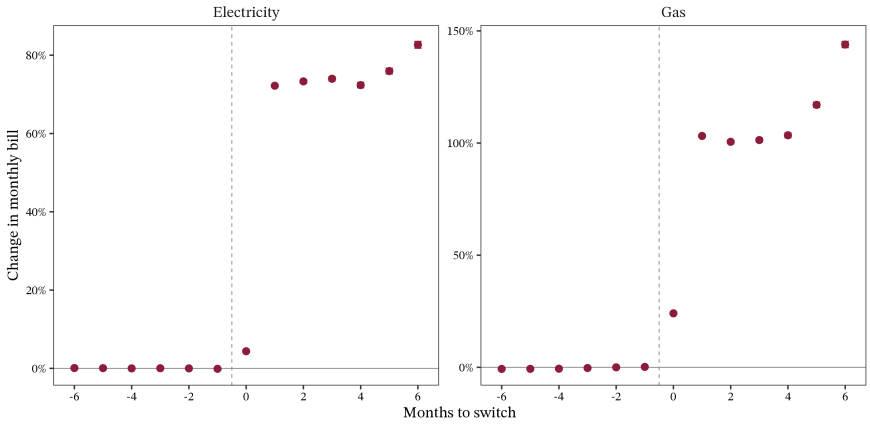
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Appendix

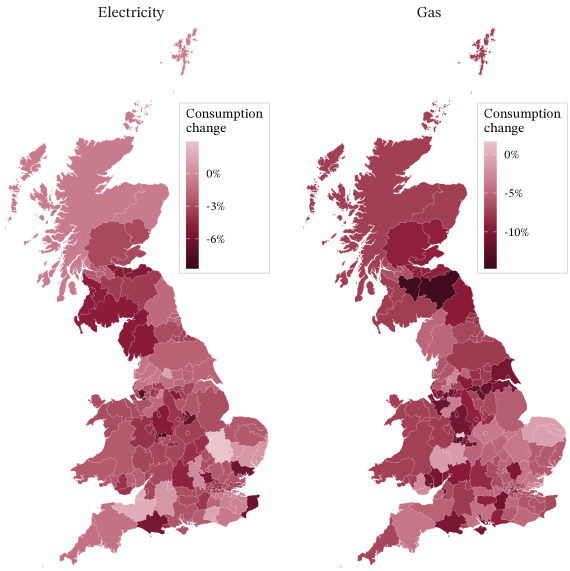
appendix: Matched to the CS comparisons. For each cohort g, compare its households with households not yet treated at g1, using data from the same calendar window, then average across cohorts with CS's cohort weights. This is the comparison your estimates actually rely on, so it's the most defensible version and worth putting in the appendix.

Figure 31: Bill Rise Event Study



Notes: This figure shows the average bill rise consumers face, including their standing daily rates and post-EPG unit rates, relative to their contract expiry date using the specification shown in Equation (2).

Figure 32: Consumption Change by Region



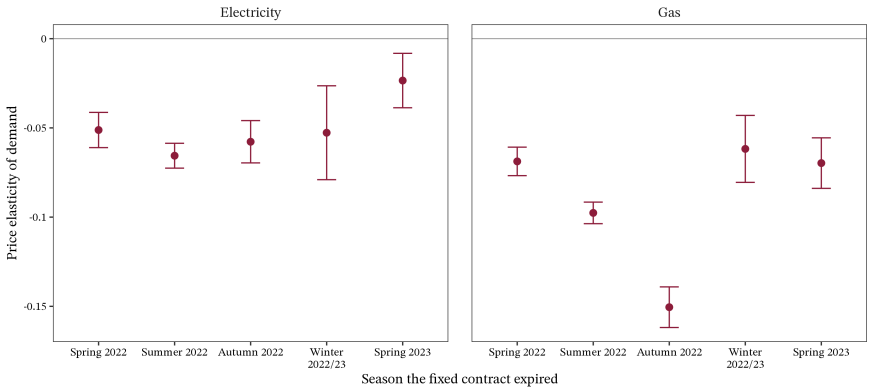
Notes: This figure shows the percentage decrease in consumption of electricity and gas by region.

Table 4: Ofgem Energy Price Cap and Government Support History

Period	Ofgem Cap	Notes / Effective Cost
1 January 2019 – 31 March 2019	£1,137	
1 April 2019 – 30 September 2019	£1,254	
1 October 2019 – 31 March 2020	£1,179	
1 April 2020 – 30 September 2020	£1,162	
1 October 2020 – 31 March 2021	£1,042	
1 April 2021 – 30 September 2021	£1,138	
1 October 2021 – 31 March 2022	£1,277	
1 April 2022 – 30 September 2022	£1,971	
1 October 2022 – 31 December 2022	£3,549	£2,500 via EPG (£2,100 with EBSS)
1 January 2023 – 31 March 2023	£4,279	£2,500 via EPG (£2,100 with EBSS)
1 April 2023 – 30 June 2023	£3,280	£2,500 via EPG
1 July 2023 – 30 September 2023	£2,074	
1 October 2023 – 31 December 2023	£1,923	
1 January 2024 – 31 March 2024	£1,928	
1 April 2024 – 30 June 2024	£1,690	
1 July 2024 – 30 September 2024	£1,568	
1 October 2024 – 31 December 2024	£1,717	
1 January 2025 – 31 March 2025	£1,738	
1 April 2025 – 30 June 2025	£1,849	
1 July 2025 – 30 September 2025	£1,720	
1 October 2025 – 31 December 2025	£1,755	
1 January 2026 – 31 March 2026	£1,758	
1 April 2026 – 30 June 2026	£1,641	
1 July 2026 – 30 September 2026	£1,862	
1 October 2026 – 31 December 2026	£1,937	

Note: Figures represent the Ofgem price cap for a typical household based on 2023 Typical Domestic Consumption Values (2,700 kWh electricity, 11,500 kWh gas, 3,900 kWh multi-register). During Winter 2022–23, the Energy Price Guarantee (EPG) capped typical bills at £2,500, with the government covering the difference. The Energy Bills Support Scheme (EBSS) provided an additional £400 government-funded discount.

Figure 33: Price Elasticity by Expiry Season



Notes: This figure displays the price elasticity of demand by season of expiry.

Table 5: Robustness Checks across Different Estimation Methods

	Electricity			Gas		
	Unit rate (%)	Consumption (%)	PED	Unit rate (%)	Consumption (%)	PED
Callaway & Sant'Anna (figure 2)	76.7	-3.07	-0.055	168.8	-7.22	-0.076
	(0.01)	(0.22)	(0.004)	(0.02)	(0.33)	(0.003)
Callaway & Sant'Anna, weather-adjusted	76.7	-3.07	-0.055	168.8	-7.22	-0.076
	(0.01)	(0.22)	(0.004)	(0.02)	(0.33)	(0.003)
Wooldridge (extended TWFE)	74.9	-2.04	-0.037	157.6	-8.20	-0.090
	(0.02)	(0.75)	(0.013)	(0.09)	(0.60)	(0.006)
Borusyak, Jaravel & Spiess	75.0	-2.47	-0.045	158.0	-7.82	-0.086
	(0.02)	(0.65)	(0.012)	(0.09)	(0.48)	(0.005)
Gardner (two-stage)	75.0	-2.47	-0.045	158.0	-7.82	-0.086
	(0.04)	(0.65)	(0.012)	(0.10)	(0.48)	(0.005)
Cengiz et al. (stacked)	92.2	-3.82	-0.060	205.1	-9.36	-0.088
	(0.11)	(0.31)	(0.005)	(0.14)	(0.46)	(0.004)
de Chaisemartin & D'Haultfœuille	78.8	-1.02	-0.018	162.5	-7.50	-0.081
	(1.35)	(1.53)	(0.026)	(1.07)	(2.24)	(0.023)

Notes: Figure 2's households (fixed contracts begun before March 2022). Effects are averages over months 1–6 after the fixed contract ended, as % changes; standard errors (log points × 100) clustered by household in parentheses. Treated cohorts: fixes ending March 2022 – April 2023; households whose fix ends later are controls only, before their own switch. All estimators compare households within the same sample (smart / non-smart, moving to a variable or a new fixed tariff), length of fix and ITL1 region, with household and month fixed effects; consumption allows one month of anticipation (base month −2) and controls for heating and cooling degree days (Callaway & Sant'Anna for figure 2: none, comparisons with at least 200 controls); the unit rate has no anticipation (base month −1). PED = consumption effect / unit-rate effect. Borusyak, Jaravel & Spiess and Gardner impute from the same model of untreated months. The stacked estimator uses a random 10% of eligible control households, with Wing, Freedman & Hollingsworth (2024) weights. Wooldridge and Gardner use all households, dropping treated household-months in region groups and months with no untreated household; Wooldridge's effects are saturated at sample × fix length × cohort × month, with month effects within region. For de Chaisemartin & D'Haultfœuille (5% random sample of households) the standard error is the mean of the six monthly standard errors, a conservative bound.

Table 6: Robustness: Varying number of controls

	Minimum controls per comparison		
	100	200 (Figure 5)	500
Panel A: Electricity			
Unit rate (%)	76.7	76.7	76.9
	(0.01)	(0.01)	(0.01)
Consumption (%)	-3.05	-3.07	-3.09
	(0.22)	(0.22)	(0.23)
Bill (%)	75.8	76.0	77.4
	(0.18)	(0.18)	(0.18)
Price elasticity	-0.054	-0.055	-0.055
	(0.004)	(0.004)	(0.004)
Treated households	136,334	130,956	114,659
Comparisons (months 1–6)	2,791	2,601	2,277
Panel B: Gas			
Unit rate (%)	168.0	168.8	171.1
	(0.02)	(0.02)	(0.02)
Consumption (%)	-7.07	-7.22	-7.53
	(0.32)	(0.33)	(0.35)
Bill (%)	115.9	116.4	116.5
	(0.23)	(0.23)	(0.25)
Price elasticity	-0.074	-0.076	-0.078
	(0.003)	(0.003)	(0.004)
Treated households	160,755	152,910	136,417
Comparisons (months 1–6)	2,929	2,690	2,352

Notes: Figure 5's households and specification (fixed contracts begun before March 2022; treated cohorts: fixes ending March 2022 – April 2023). Callaway & Sant'Anna with not-yet-expired controls in the same sample (smart / non-smart, moving to a variable or a new fixed tariff), length of fix and ITL1 region. Each column drops any comparison (expiry cohort × group × month) with fewer control households than the minimum. Effects are averages over months 1–6 after the fixed contract ended, as % changes.

.1 Optimization Lagrangian

The consumer solves the following optimization problem:

$$\max_{C,E} \quad \mathcal{L}(C,E,\lambda) = C^{1-\alpha}\left(E-\overline{E}\right)^{\alpha} + \lambda\left[Y+S-C-(p-\sigma)E\right] \tag{18}$$

The First-Order Conditions (FOCs) with respect to C, E, and λ are:

$$\frac{\partial \mathcal{L}}{\partial C} = (1-\alpha)C^{-\alpha}\left(E-\overline{E}\right)^{\alpha} - \lambda = 0 \tag{19}$$

$$\frac{\partial \mathcal{L}}{\partial E} = \alpha C^{1-\alpha}\left(E-\overline{E}\right)^{\alpha-1} - \lambda(p-\sigma) = 0 \tag{20}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = Y+S-C-(p-\sigma)E = 0 \tag{21}$$

Dividing Equation (20) by Equation (19) yields the Marginal Rate of Substitution (MRS):

$$\text{MRS}_{E,C} = \frac{\alpha}{1-\alpha}\cdot\frac{C}{E-\overline{E}} = p-\sigma \implies C = \frac{1-\alpha}{\alpha}(p-\sigma)(E-\overline{E}) \tag{22}$$

Substituting Equation (22) into the budget constraint (21):

$$\left[\frac{1-\alpha}{\alpha}(p-\sigma)(E-\overline{E})\right] + (p-\sigma)E = Y+S \tag{23}$$

Expanding and isolating optimal electricity consumption E^*:

$$(p-\sigma)E\left(\frac{1-\alpha}{\alpha}+1\right) - \frac{1-\alpha}{\alpha}(p-\sigma)\overline{E} = Y+S \tag{24}$$

$$\frac{p-\sigma}{\alpha}E = Y+S+\frac{1-\alpha}{\alpha}(p-\sigma)\overline{E} \tag{25}$$

$$E^*(p,\sigma,S) = \overline{E} + \frac{\alpha}{p-\sigma}\left[Y+S-(p-\sigma)\overline{E}\right] \tag{26}$$